

Basal forebrain nuclei involvement in post-stroke aphasia: Combined effects of cholinergic stimulation and language therapy

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BACKGROUND AND OBJECTIVES

Although the acetylcholinesterase inhibitor donepezil is the most extensively studied agent for the treatment of post-stroke aphasia (PSA), the confirmation of cholinergic depletion has traditionally been challenging, leading to neurochemical diagnostic and drug selection uncertainty.¹

Therefore, identifying damage to the nucleus basalis of Meynert (Ch4), the primary source of acetylcholine projections to the anterior language region, clearly justifies using cholinergic agents to treat PSA.

We examined the involvement of the cholinergic basal forebrain (CBF) sectors in patients with chronic PSA and the therapeutic effects of combining cholinergic stimulation with intensive language-action therapy (ILAT).^{1,2}

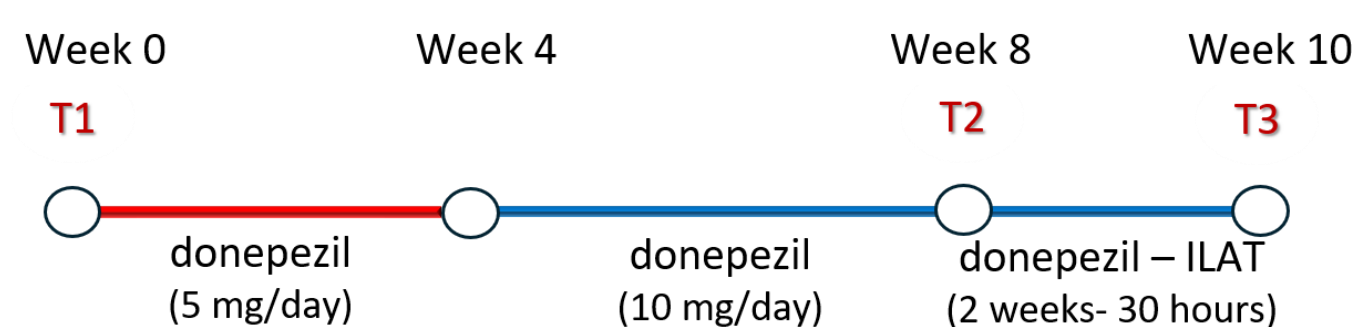
METHODS

Participants: 30 patients with chronic PSA (≥ 6 months post-onset) of mild-to-moderate severity, who had chronic left hemisphere lesions.

Testing: Western Aphasia Battery – Revised (WAB-R).

Treatment responder criterion: Participants who improved ≥ 5 points on the WAB-R Aphasia Quotient were classified as treatment responders.

Study Design



Neuroimaging data analysis: We performed longitudinal voxel-based morphometry (VBM) analyses of grey matter (GM) using SPM12 on serial 3T T1-weighted MRI scans acquired at different time points. We classified responder participants by left CFN status (CBF-lesioned; CBF-spared) and performed one-sample t-tests and correlations with WAB-R scores in SPM12 ($p < 0.001$ voxel-level; $p < 0.05$ FWE cluster-corrected). The study is registered with EudraCT (2008–008481-12) and NCT04134416.

CONCLUSIONS

Our findings provide evidence of two distinct spatial locations and phases of adaptive structural plasticity (map expansion and renormalization)^{3,4} in chronic aphasic patients who responded to combined donepezil and intensive training, primarily in those with damage to the left posterior Ch4.

REFERENCES

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ACKNOWLEDGEMENTS

Special thanks to Dr. Grothe M. for sharing his BF mask.

RESULTS

Baseline (T1) Lesion Characteristics

Figure 1. Lesion Overlap Map

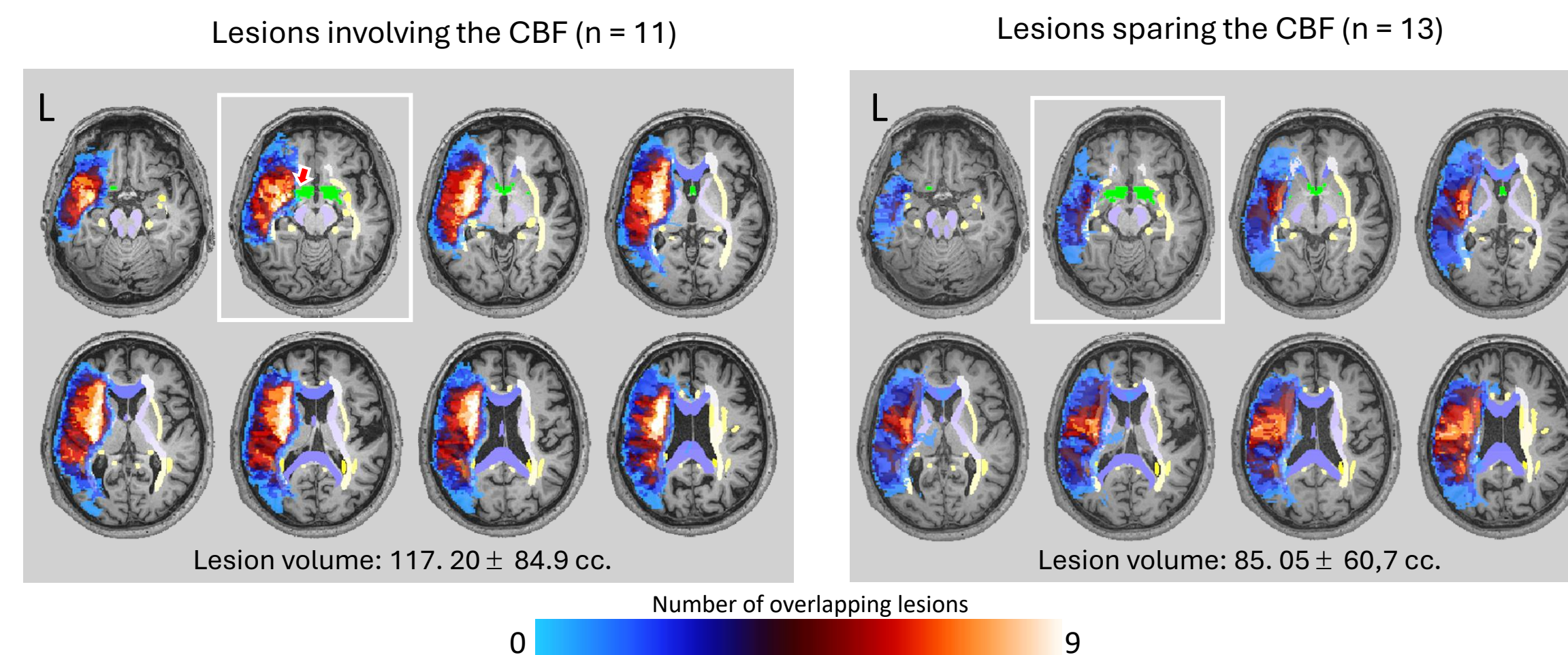
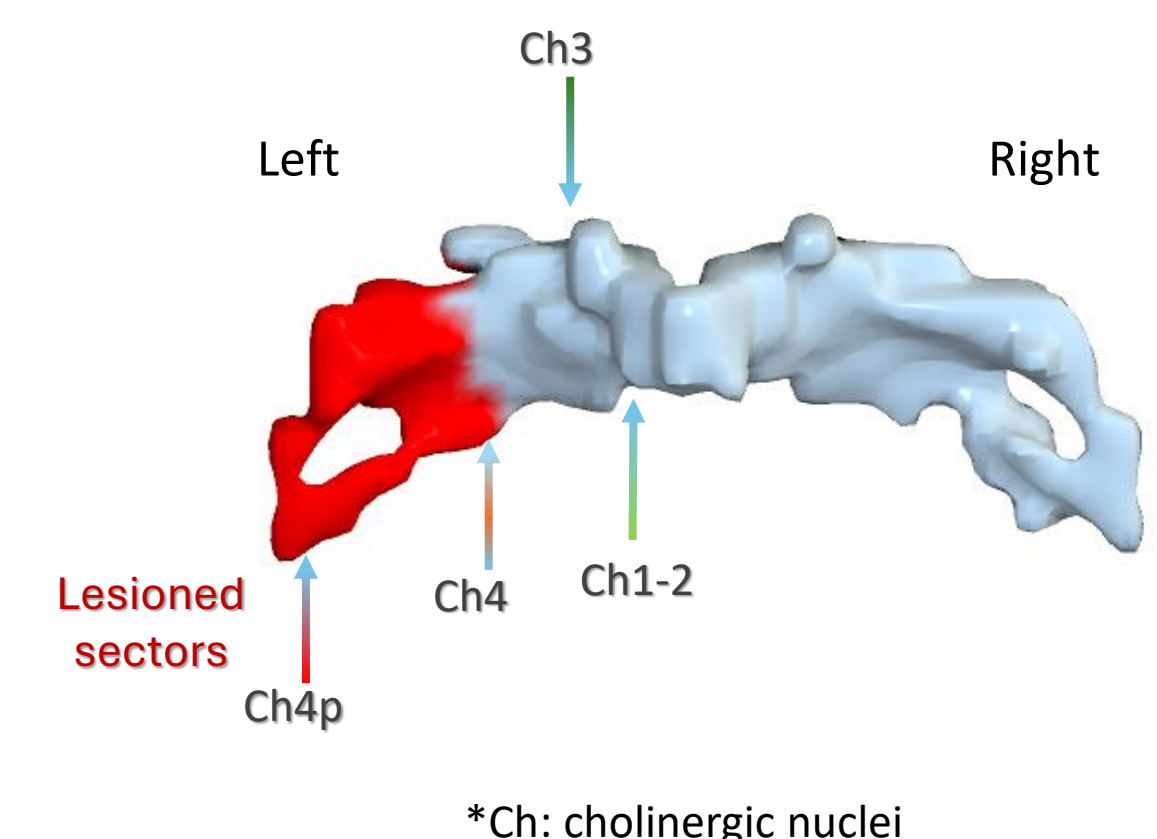


Figure 2. Baseline CBF Mask Ch1-4 subsectors*



Donepezil alone (T2) and combined with ILAT (T3) results

Figure 3. WAB-R Aphasia Quotient, Mean Change from Baseline (Δ)

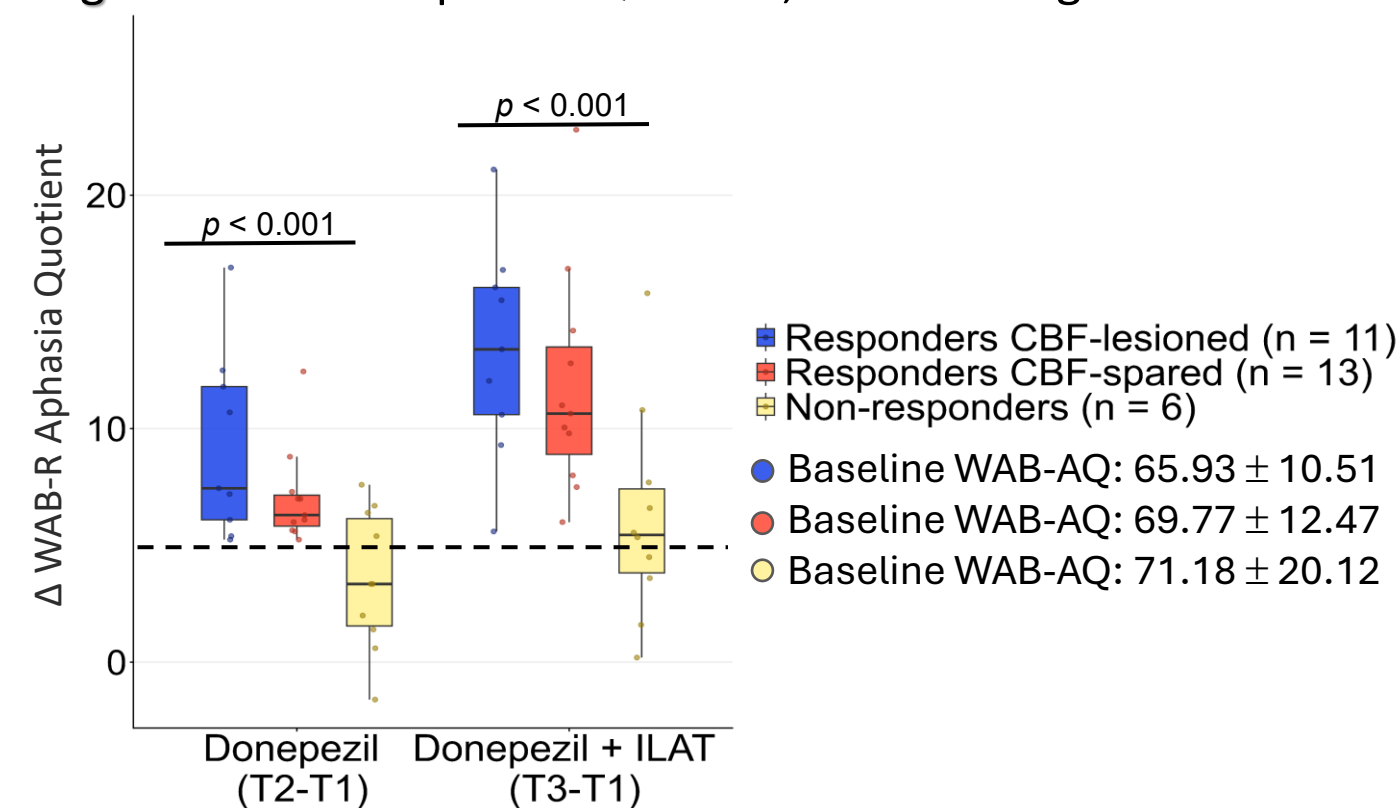


Figure 4. Right Caudate-Putamen Expansion under Donepezil alone (T2-T1)

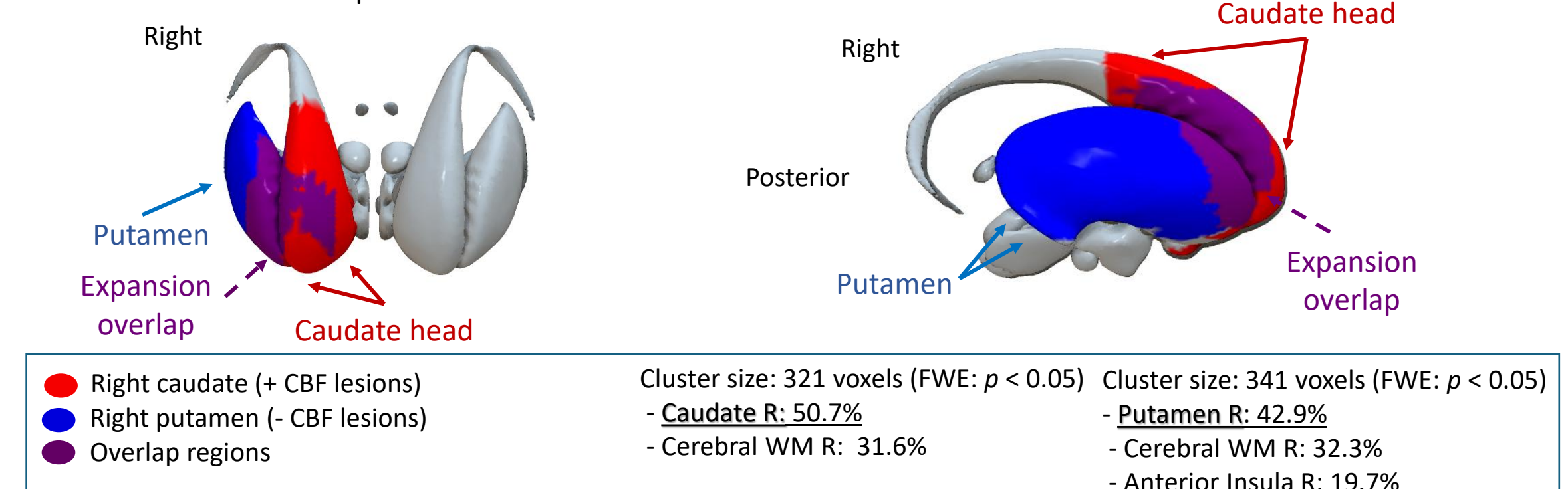
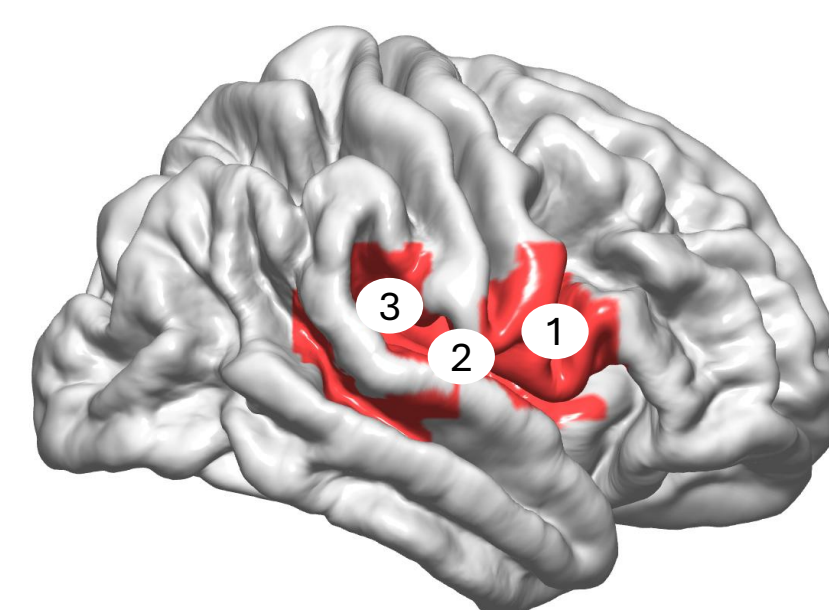


Figure 5. Right Cortical Map Plasticity after Donepezil-ILAT (T3-T1)



- T3-T1 GMV changes correlated positively with WAB Spontaneous Speech scores only in the CBF-lesioned group: (pFWE-corrected at cluster-level = 2.31×10^{-8} , $k = 1553$ voxels).
- (1) Central operculum
- (2) Transverse temporal gyrus
- (3) Posterior insula
- No volumetric changes were observed in the right CBF nor in the left hemisphere.

Map Expansion^{3,4}

- In both groups VBM-GMV changes after donepezil alone (T2-T1, week 8) did not correlate with language performance.

Renormalization^{3,4}

- The initial GMV striatum expansion (T2-T1, week 8) returned to baseline volume after adding ILAT to donepezil (T3-T1, week 10).